1. TCP Communication Flag Types

TCP uses flags (control bits) in the header to manage communication states. The key flags are:

- **SYN (Synchronize):** Initiates a TCP connection (start of handshake).
- **ACK (Acknowledgment):** Acknowledges received data.
- **FIN (Finish):** Gracefully ends a connection.
- **RST (Reset):** Abruptly terminates a connection.
- **PSH (Push):** Requests immediate data processing.
- **URG (Urgent):** Marks urgent data in the segment.

Flags control connection establishment, data transfer, and termination.

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2. Banner Grabbing and OS Fingerprinting Techniques

- **Banner Grabbing:** Collecting info about services running on open ports by connecting and reading the welcome messages (banners) sent by servers. This helps identify software and versions.
- **OS Fingerprinting:** Determines the operating system of a target by analyzing responses to crafted network packets, TCP/IP stack behavior, or subtle differences in protocols.

Two types:

- **Active fingerprinting:** Sending packets and analyzing responses.
- **Passive fingerprinting:** Monitoring network traffic without interaction.

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3. How Proxy Servers Are Used in Launching Attacks

Attackers use proxy servers to hide their IP addresses and anonymize attacks, making tracing difficult. Proxies can also relay malicious requests or serve as intermediate hosts in attacks like:

- **Proxy chaining:** Using multiple proxies to increase anonymity.
- **Open proxies:** Misconfigured or public proxies abused for attacks.
- **Anonymizing attacks:** Hiding origin during scanning, brute force, or DoS.

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4. HTTP Tunneling Techniques

HTTP tunneling encapsulates non-HTTP traffic inside HTTP requests and responses to bypass firewalls or proxies that only allow HTTP/S traffic. This enables attackers to:

- Establish covert channels.
- Evade security devices.

- Carry out command and control communication over allowed HTTP ports (usually 80 or 443).

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5. IP Spoofing Techniques

IP spoofing involves forging the source IP address in IP packets to impersonate another system. Goals include:

- Concealing attack origin.
- Bypassing IP-based authentication.
- Launching DoS or reflection attacks.

Techniques:

- **Blind spoofing:** Attacker sends packets without expecting a reply.
- **Non-blind spoofing:** Attacker predicts sequence numbers and intercepts replies.

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6. Enumeration

Enumeration is the process of extracting detailed info from a system/network, such as:

- Usernames

- Group info
- Network shares
- Services running

It often follows footprinting and scanning, forming the basis for later attacks.

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7. Password-Cracking Techniques

- **Brute Force:** Trying all possible combinations.
- **Dictionary Attack:** Using a precompiled list of probable passwords.
- **Hybrid Attack:** Combines dictionary with mutations (e.g., adding numbers).
- **Rainbow Tables:** Precomputed hash values to reverse hashes quickly.
- **Credential Stuffing:** Using leaked passwords from one system on others.

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8. Cracking Windows Passwords

Windows stores password hashes in the Security Accounts Manager (SAM) file. Attackers:

- Extract hashes using tools like pwdump.
- Use cracking tools like John the Ripper or Hashcat.
- Exploit weak passwords or outdated hashing algorithms (LM hash).

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9. Redirecting the SMB Logon to the Attackers

SMB (Server Message Block) protocol is used for file sharing. Attackers redirect SMB authentication requests to their machines to capture credentials, often via:

- **SMB Relay attacks:** Intercept and forward authentication requests.
- **Man-in-the-middle (MITM) attacks:** Intercept traffic between client and server.

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10. SMB Redirection, SMB Relay MITM Attacks and Countermeasures

- **SMB Relay Attack:** Attacker relays SMB authentication messages to gain unauthorized access.
- **Countermeasures:** Use SMB signing, enforce NTLMv2, disable NTLMv1, patch systems, and monitor network traffic.

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11. NetBIOS DoS Attacks

NetBIOS is a network protocol on Windows. Attackers can flood NetBIOS services with requests to overwhelm and cause denial of service.

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12. DDoS Attack

Distributed Denial of Service (DDoS) attacks overwhelm a target system/network with massive traffic from multiple sources, causing service disruption.

Types:

- **Volumetric attacks:** Flood bandwidth.
- **Protocol attacks:** Exploit protocol weaknesses.
- **Application layer attacks:** Target specific services.